

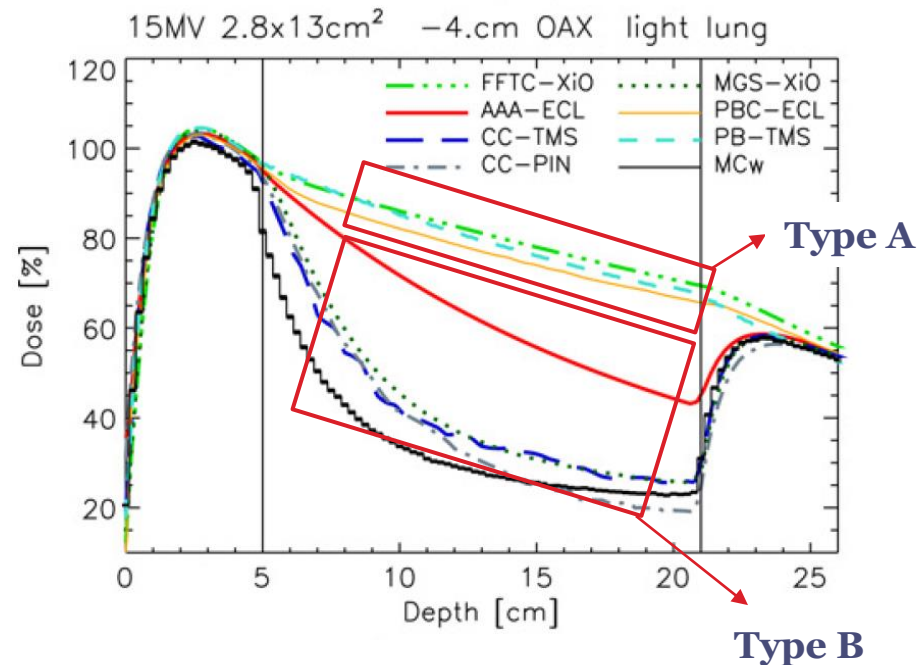
Dose Computation Algorithms

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September 30th, 2016

Basic Treatment Planning Rotation -
Session 8

Type A vs. Type B

- **Type A** – Models primarily based on EPL scaling for inhomogeneity corrections.
 - Eclipse/SPB, OMP/PB, PPLAN, XiO/Convolution
 - **LONGITUDINAL** scaling
- **Type B** – Models that in an approximate way consider changes in lateral electron transport
 - Pinnacle/CC, Eclipse/AAA, OMP/CC, XiO/Superpositioning
 - **LONGITUDINAL and LATERAL** scaling



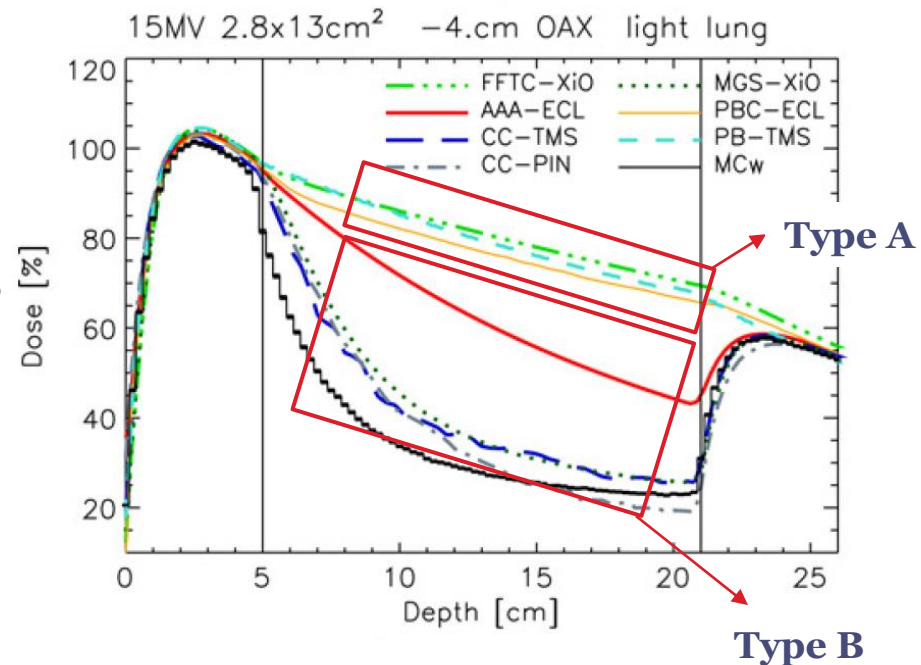
Type A vs. Type B

- **Type A**

- Do not model electron energy transport
- Inadequate in heterogeneous materials (esp for small fields)

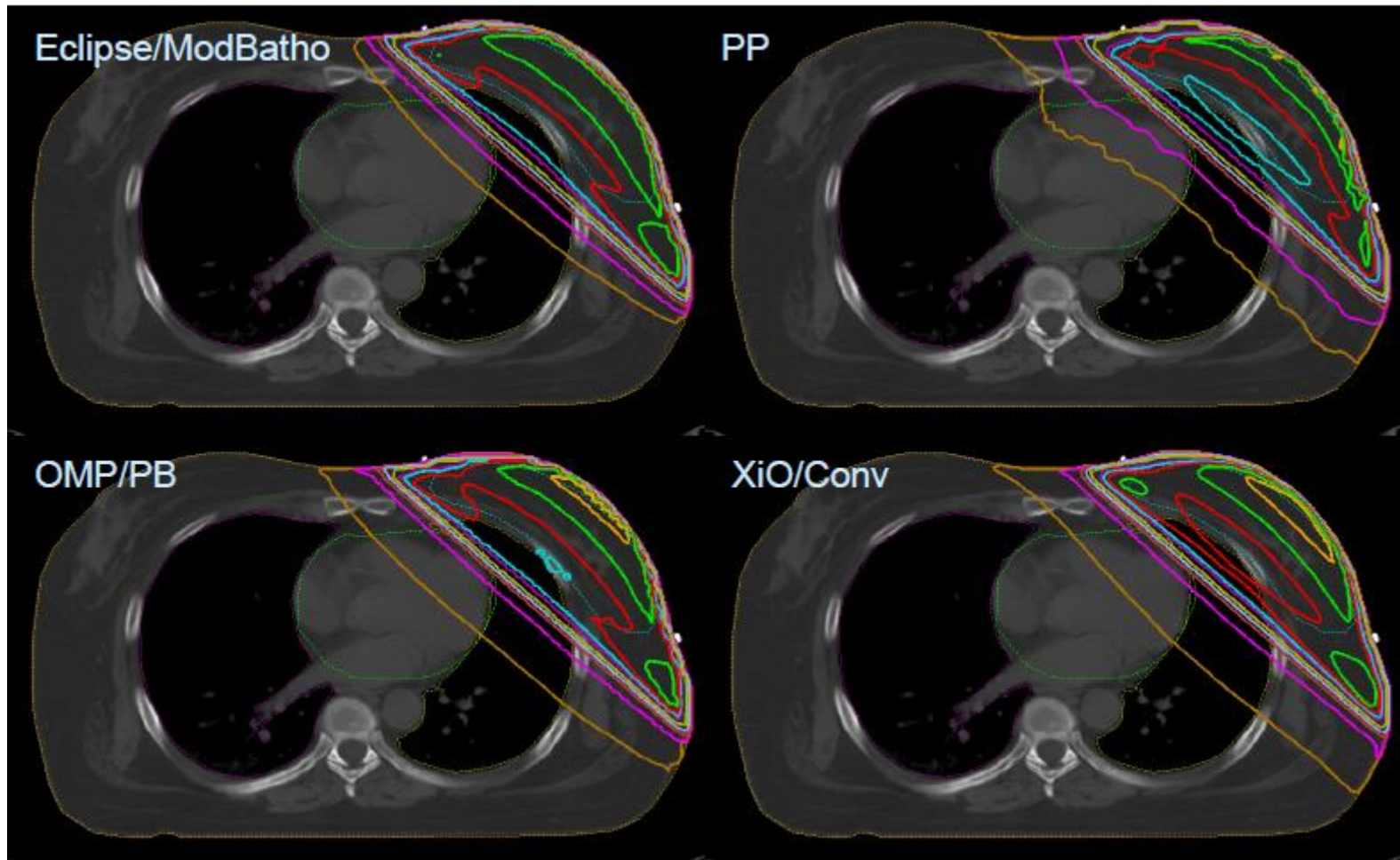
- **Type B**

- Pinnacle (CC) handles electronic disequilibrium through lung better than Eclipse (AAA)



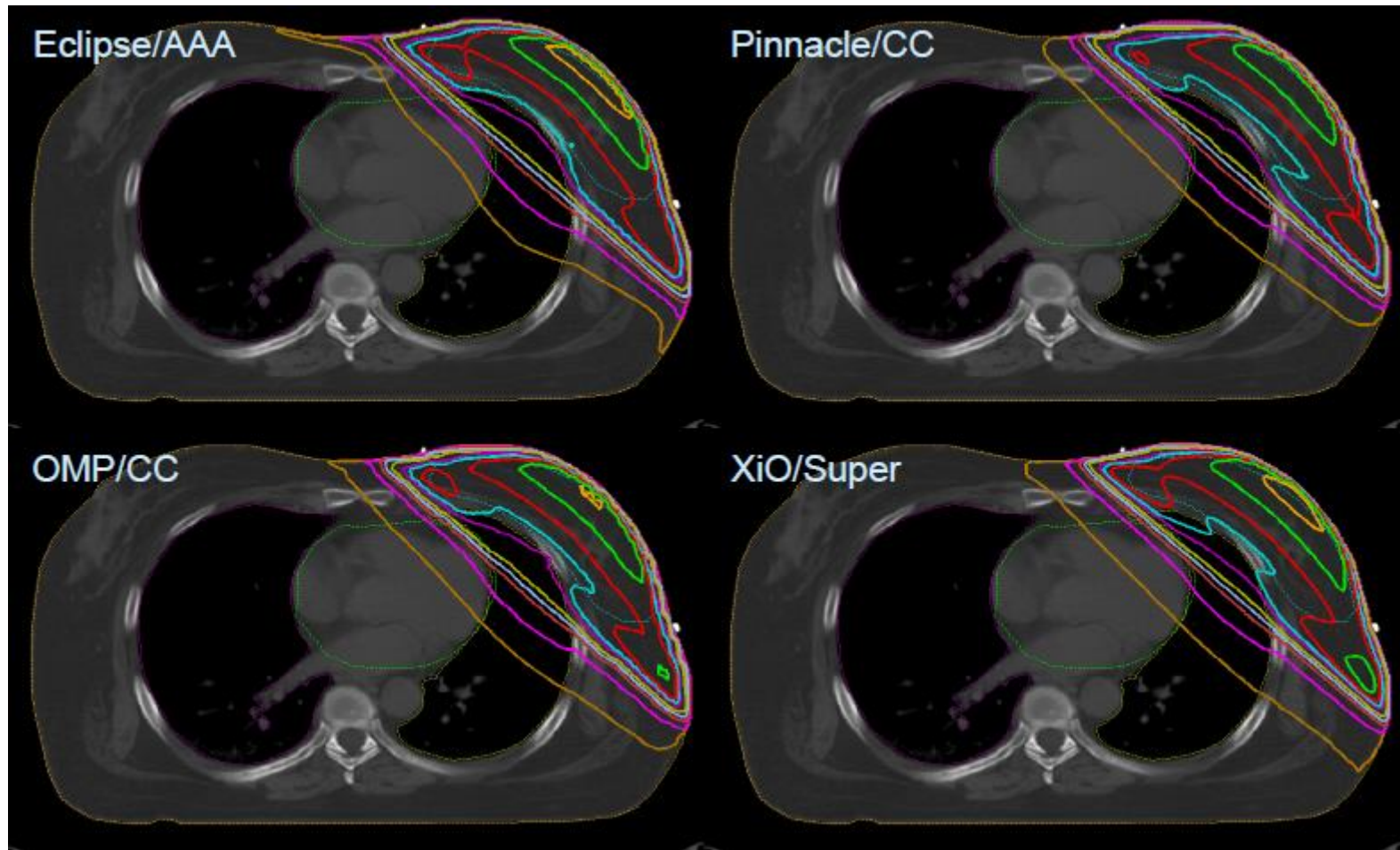
Clinical Case Comparison: Breast 6 MV, Type A

Breast 6 MV, Type A



Clinical Case Comparison: Breast 6 MV, Type B

Breast 6 MV, Type B



Comparing Clinical Cases

Dose Statistics for Breast

Type A

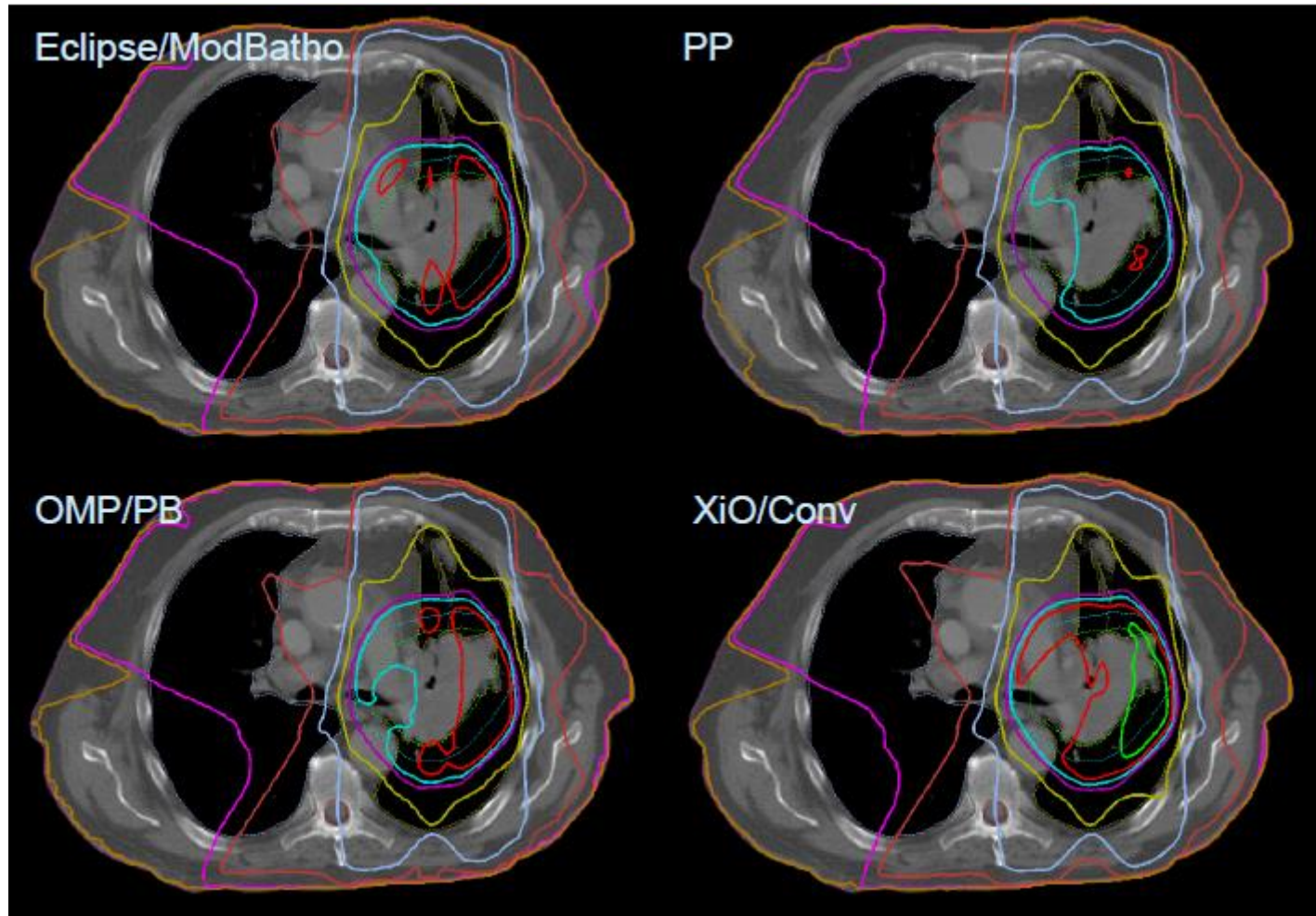
Type B

6 MV	Eclipse/ Mod Batho 2	OMP/PB	PP	Xio/Conv	Eclipse/ AAA 2	OPM/CC	Pinnacle/CC	Xio/ Super
PTV Mean	101.5	100.3	97.4	101.7	100.3	98.9	97.6	100.3
PTV D ₉₅	93.0	91.4	88.1	93.0	91.4	89.8	88.9	91.4
PTV D ₅	109.1	109.1	106.7	111.5	110.7	108.3	106.7	109.9
PTV D ₅ -D ₉₅	16.1	17.7	18.5	18.5	19.3	18.5	17.7	18.5
Pulm sin D ₅	91.4	93.8	89.8	95.4	83.3	82.5	80.9	85.7
Pulm sin D ₅₀	2.8	3.6	3.6	4.4	4.4	3.6	3.6	4.4
15/18 MV	Eclipse/ Mod Batho 2	OMP/PB	PP	Xio/Conv	Eclipse/ AAA 2	OPM/CC	Pinnacle/CC	Xio/ Super
PTV Mean	101.3	100.2	97.5	101.5	99.9	96.3	97.5	100.2
PTV D ₉₅	95.4	93.8	92.1	95.4	93.8	90.5	90.5	94.6
PTV D ₅	106.8	106.8	101.9	108.4	106.8	101.9	103.5	107.6
PTV D ₅ -D ₉₅	11.4	13.0	9.7	13.0	13.0	11.4	13.0	13.0
Pulm sin D ₅	92.1	96.2	91.3	97.0	84.8	80.8	79.2	84.0
Pulm sin D ₅₀	2.0	2.8	5.3	3.7	2.8	3.7	2.8	3.7

The average PTV dose decreases with 0.7% and 1.6% for 6 and 15/18 MV, respectively, when replacing *type a* with *type b* models. For high dose region in the adjacent left lung, D₅ decreases significantly for both energies, for 6 MV with 9.5% and for 15/18 MV with 12.2%.

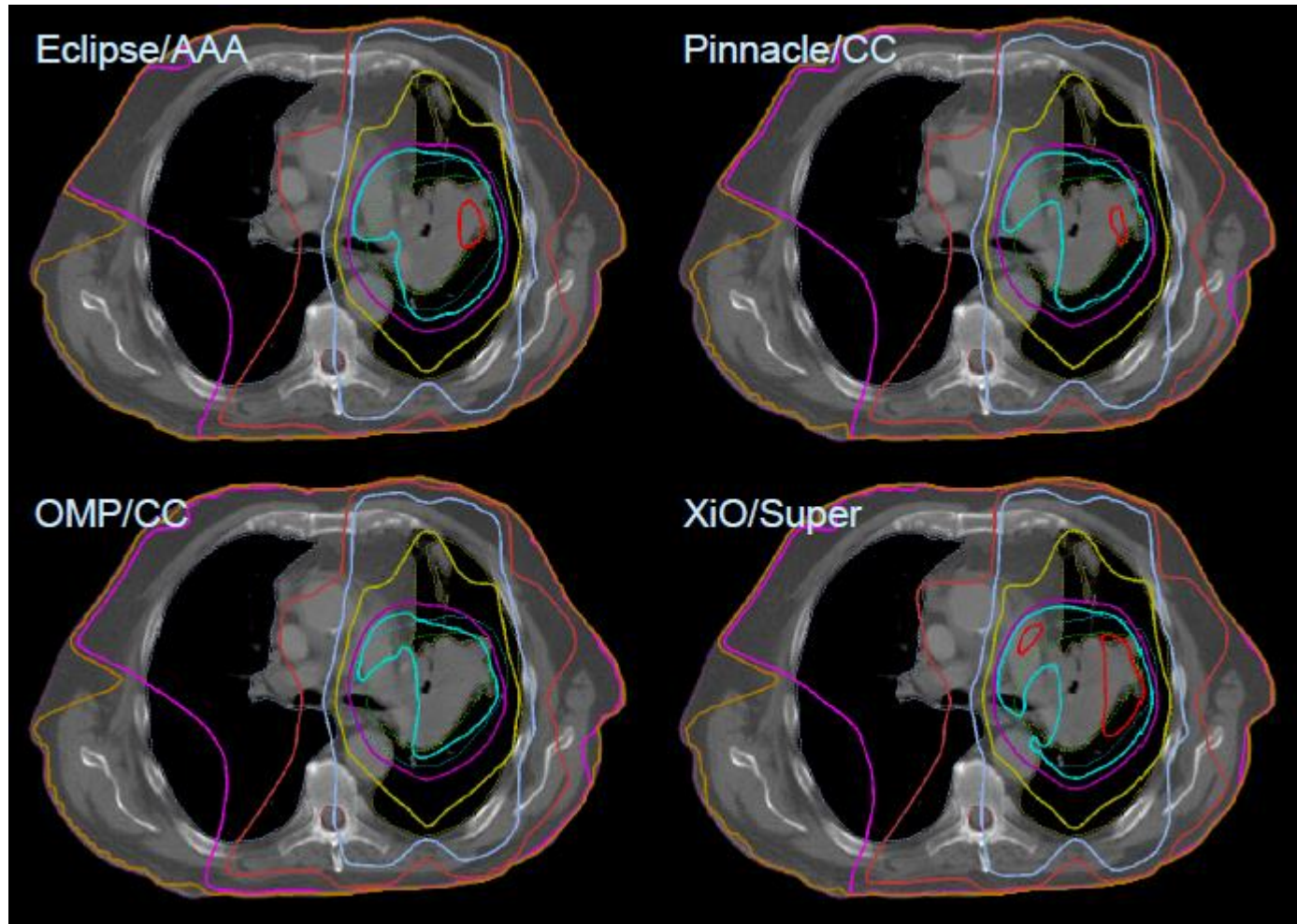
Clinical Case Comparison: Lung 18 MV, Type A

Lung 18 MV, Type A



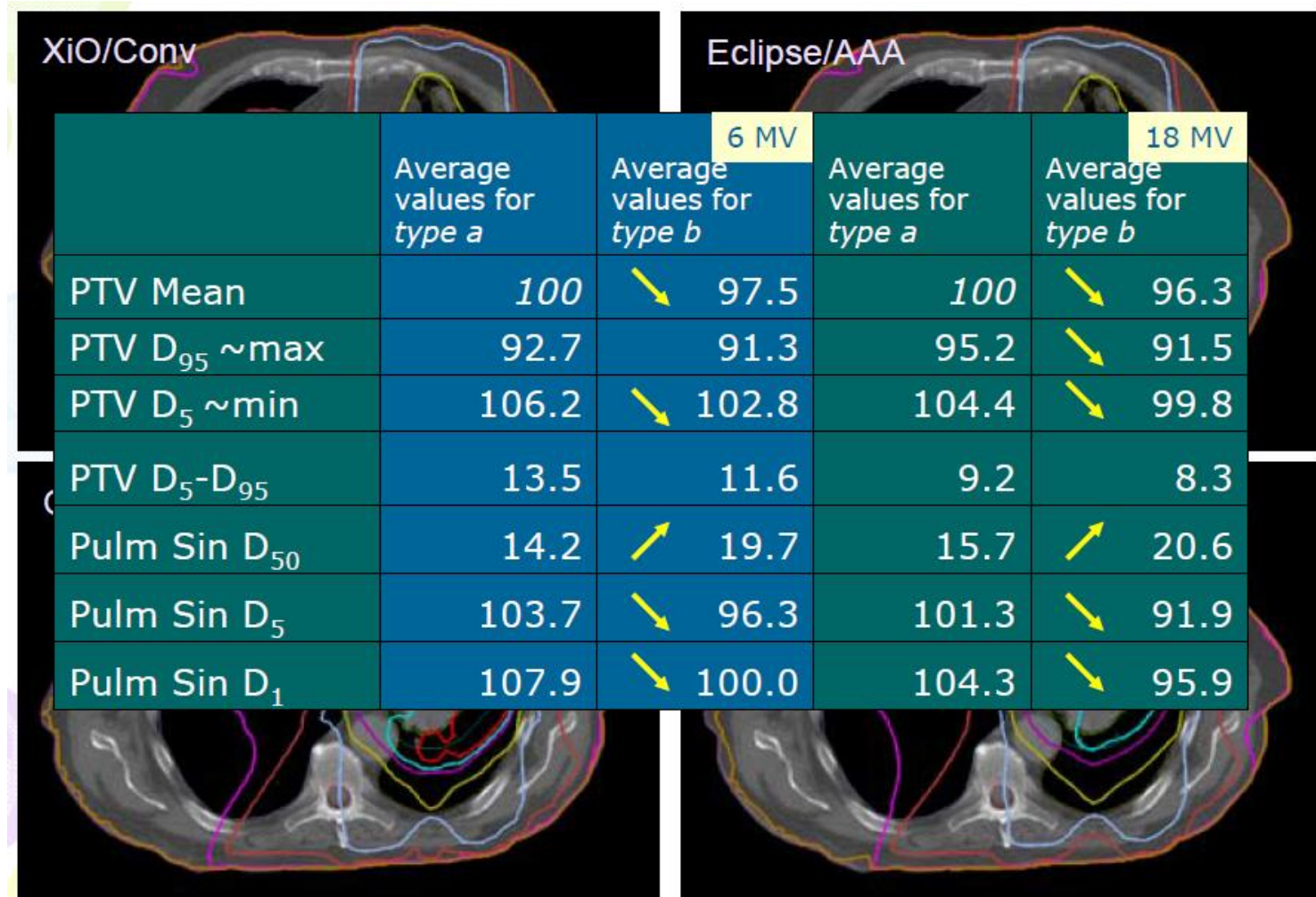
Clinical Case Comparison: Lung 18 MV, Type B

Lung 18 MV, Type B



Comparing Clinical Cases

Dose Statistics for Lung

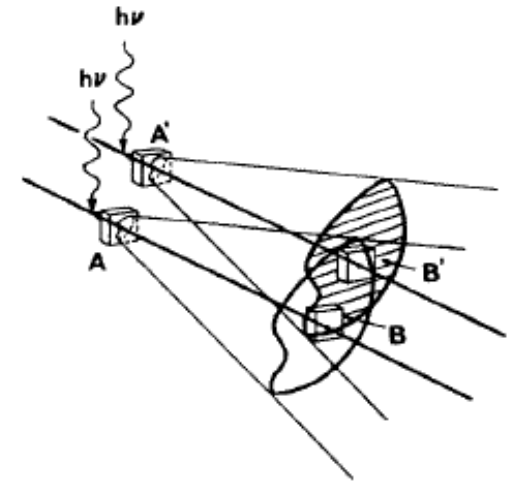


Algorithm comparisons for clinical cases

- Prostate/Pelvis planning: A or B sufficient
- Thoracic/Head & Neck – type B recommended
- Type B generally more accurate in the absence of electronic equilibrium

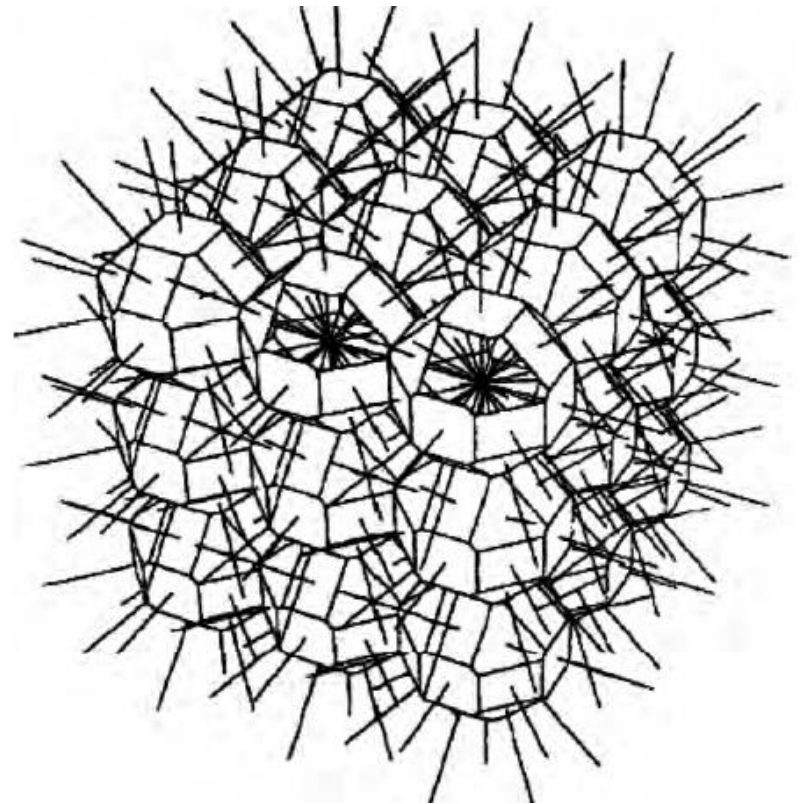
Collapsed Cone Convolution:

- Angular discretized kernel is convolved with the terma distribution
- All energy released from primary photons at elements on an axis of direction is rectilinearly transported and deposited on the axis.
- Energy that should be deposited in voxel B' from interactions at the vertex of the lower cone is deposited in voxel B and vice versa.
- Approximation is less accurate at large distances from cone vertex.
- Errors are small due to rapid fall-off of point-spread functions



Collapsed Cone Convolution:

- If we draw a 3D set of m lines, there will be nm voxels sampled for each of the n^3 deposition voxels reducing number of operations from n^3 to n^2m
 - Saves time
- CC considers density effect on kernel



CCC

- The distribution of primary energy is convolved with a poly-energetic weighted kernels that describe the energy spread of secondary particles and accounts for the lateral transport of energy
 - Uses a multiple source model to represent the clinical beam; TERMA separated into primary and scatter
 - The volume around the interaction site is discretized in 106 different directions
 - Primary and scatter transported separately along 106 lines from interaction point
 - Energy for each voxel intersected by fan line in the irradiated medium is collected and deposited according to elemental composition of medium and the density variations along the fan line

Adaptive CCC (Pinnacle)

- Only performs convolution at every 4th point in the TERMA array
- Gradient search performed on TERMA array
- Dose in between is interpolated if gradient low (i.e TERMA doesn't change much)
- Convolution performed at every point if TERMA gradient high

CCC

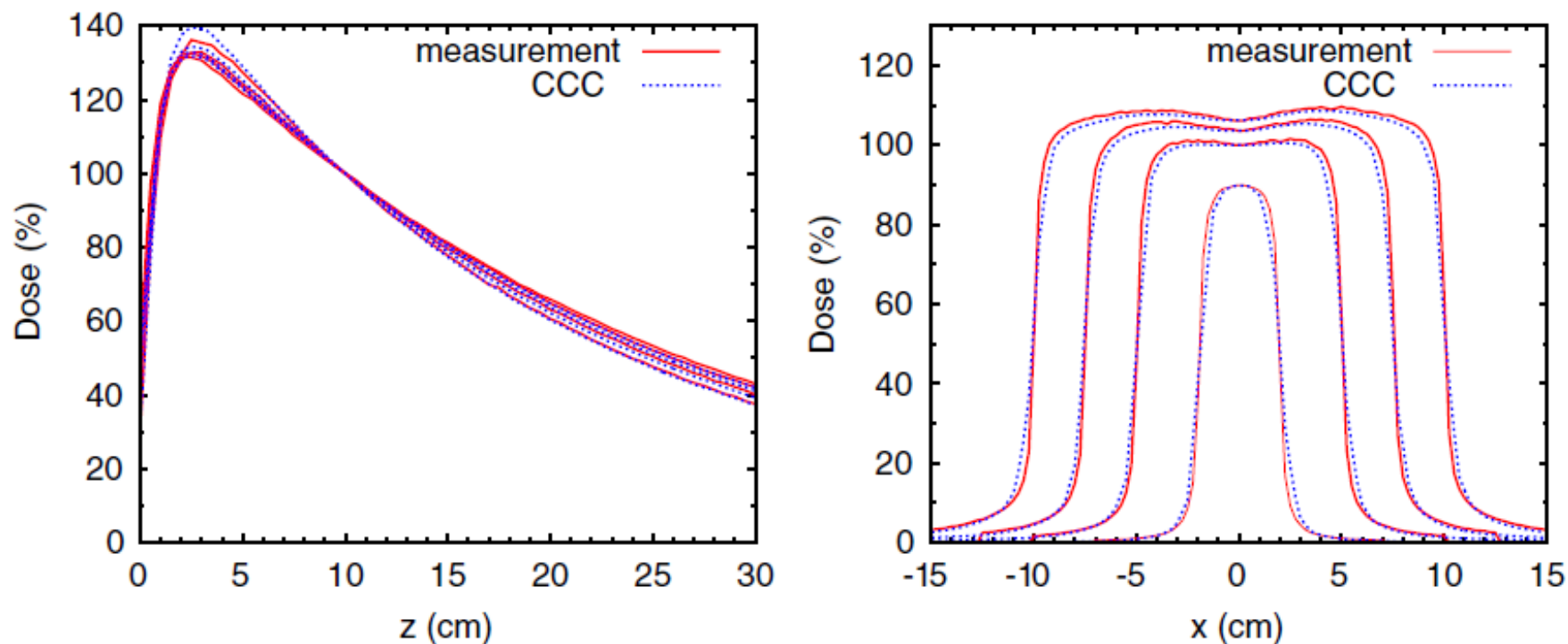
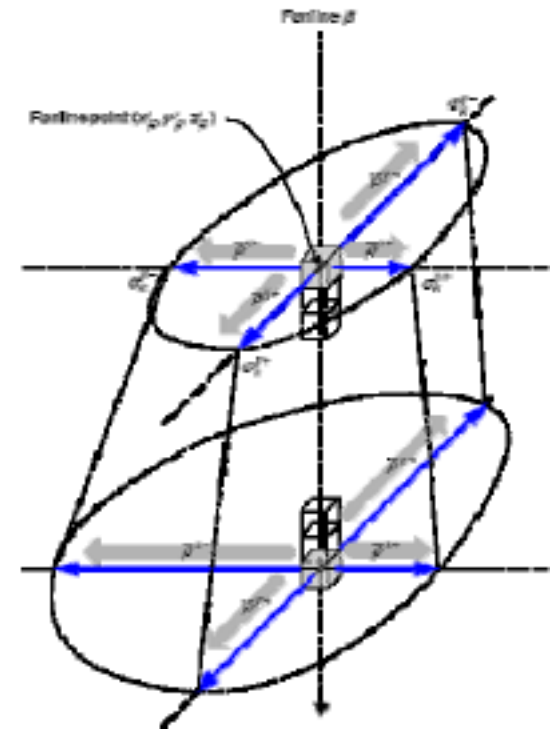


Figure 2. 15 MV percentage depth dose curves (left) and profiles at $z = 10$ cm (right) for field sizes of 4×4 , 10×10 , 15×15 and 20×20 cm². Depth dose curves are normalized to 100% at $z = 10$ cm and profiles to the corresponding output factors.

Anisotropic Analytical Algorithm (AAA) in Eclipse

- ‘anisotropic’ – considers the heterogeneity
- ‘analytical’ – it uses an analytical method, not FFT
- Pencil beams include separately modelled contributions for 3 sources
 - Primary photons
 - Extra-focal photons
 - Contaminating electrons
 - Each source has an associated fluence, energy density deposition function, and a scatter kernel (spatially invariant, MC derived)
 - Lateral energy scatter kernels are scaled anisotropically according to local electron density
 - Energy deposition functions undergo density scaling in direction (EPL) of the pencil beam (1D)

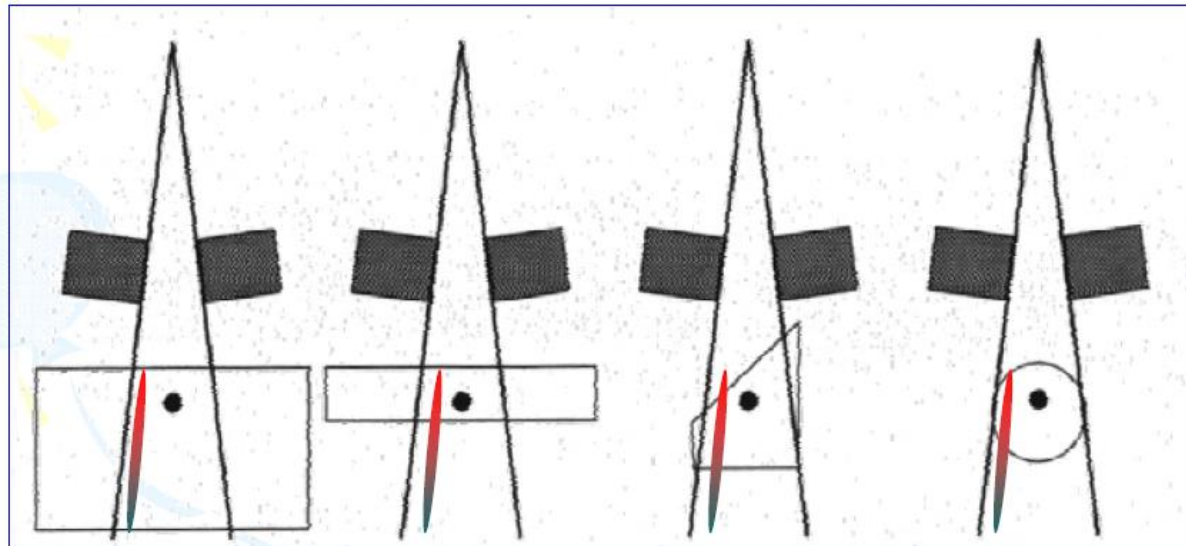


Anisotropic Analytical Algorithm (AAA) in Eclipse

- Total Energy for each pencil beam is obtained by convolving these 3 sources and final dose is calculated by superposition of contributions from pencil beams
- Clinical beam: divided into beamlets, whose cross sections are related to the calculation grid size
 - Makes assumption that the dose at a given pt is the sum of contributions of a depth dependent and a lateral part, $\perp$ to the incident direction of the beamlet
 - Affect accuracy of heterogeneity correction since it does not take into account divergent scatter from upper levels correctly
 - All depth dependent functions computed along the central fan line of the beamlet
 - Lateral energy transport is modelled by photon and electron scatter kernels discretized in 16 lateral directions

Anisotropic Analytical Algorithm (AAA) in Eclipse

- ‘anisotropic’ – considers the heterogeneity
- ‘analytical’ – it uses an analytical method, not FFT
- Clinical beam: divided into beamlets, whose cross sections are related



The same dose to ● in all geometries since the PB is pre-integrated to a certain depth/length

- Makes contribution to the incident dose
 - Affected by the level of the central
- All depend on the fan line
 - Lateral energy transport is modelled by photon and electron scatter kernels discretized in 16 lateral directions

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 the central

Anisotropic Analytical Algorithm (AAA) in Eclipse

- Heterogeneities accounted for by scaling the beamlet attenuation function by ratio of electron density of the medium to electron density of water and using equivalent depth (longitudinal)
- Lateral energy kernel scaled by equivalent path (lateral)
 - Adds a scaling of the spread of the pencil based on density
- Due to discrete number of angular sectors used a smoothing out of calculated dose distributions occurs near heterogeneities
- Uses only analytic functions which reduces computational time considerably

Anisotropic Analytical Algorithm (AAA) in Eclipse

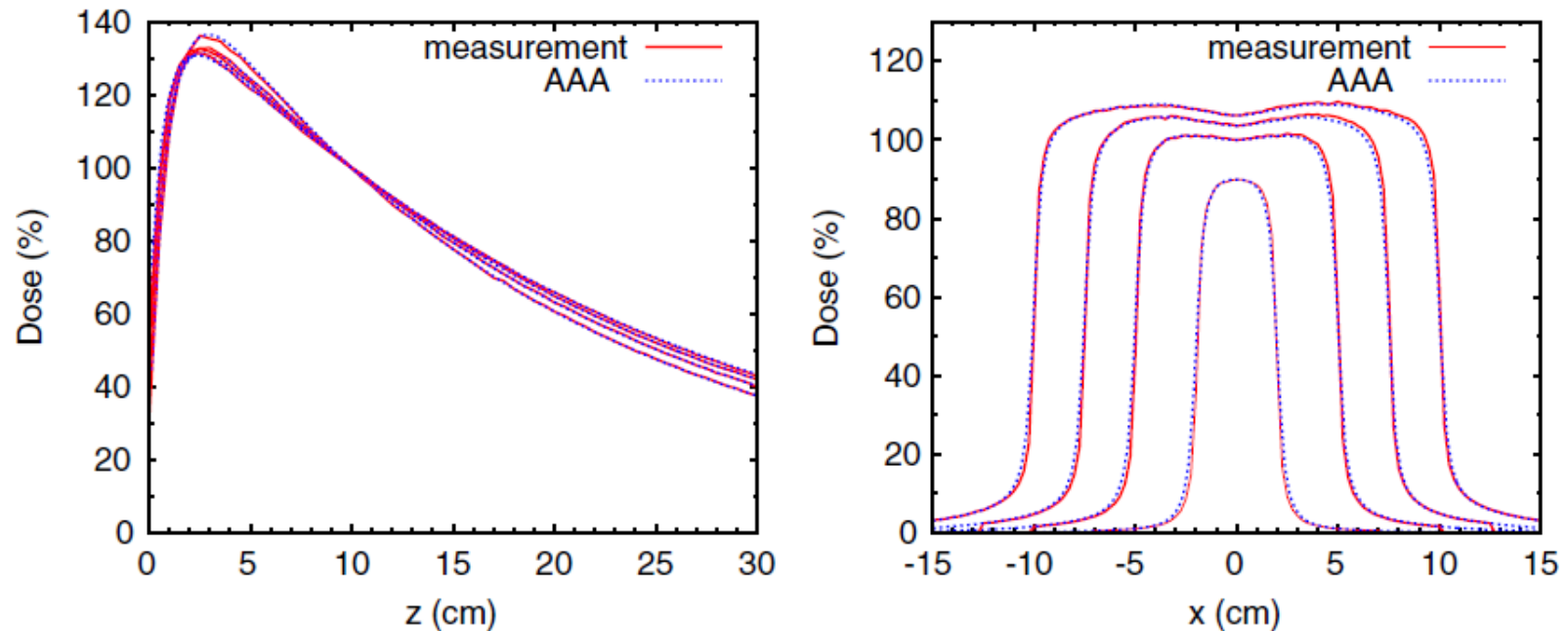


Figure 3. 15 MV percentage depth dose curves (left) and profiles at $z = 10$ cm (right) for field sizes of 4×4 , 10×10 , 15×15 and 20×20 cm². Depth dose curves are normalized to 100% at $z = 10$ cm and profiles to the corresponding output factors.